

Mauro Alejandro Jimenez Medina

MSc Electronics Engineer and Sr Data Engineer

 Barranquilla, Colombia

 mauroalejandroj

 jmauro@uninorte.edu.co

SKILLS	Machine Learning	NumpPy, Pandas, Scikit-Learn, Keras, PyTorch, NLTK, Weka, Google Colab.
	Databases	MongoDB, PostgreSQL, MySQL, MariaDB, DynamoDB, SQLite, Turso.
	Programming and Tools	Python3, Javascript (ES6), Node.js, Nuxt.js, Vue.js, C++, C#, VHDL, Spark, Arrow, Scala, Git, Docker, Google Cloud, AWS, Bitbucket, Agile (Scrum).
	Languages	Spanish (Native) and English (ITEP).
EXPERIENCE PROFESSIONAL	DexFreight - Sr Data Engineer	08/2019 - Present
	Barranquilla, Atlántico	
	• Technical Leadership & Project Management: Led the end-to-end lifecycle of port operations optimization initiatives. Managed multidisciplinary teams and coordinated agile (Scrum) sprints, translating complex business requirements into data architectures and Machine Learning models, ensuring successful production deployment on azure cloud and reducing operational bottlenecks. • Deep Learning Models (NLP/Audio): Developed a VMRS decoding system employing a speech-to-text transformer model implemented with Python and PyTorch (Keras for logs), optimized to process audio samples with a signal-to-noise ratio (SNR) over -58dB. • Operations Optimization: Designed a fleet management algorithm based on state machines using Python, NumPy, and Pandas. This optimized trucking operations, reducing driver underutilization by 5%, deadhead miles by 27%, and non-revenue hours by 44%. • Routing & Prediction Algorithms: Built a trip planning algorithm combining geospatial features and Bayes' theorem via Python and Pandas, processing thousands of records with an average response time under 5 seconds. • Recommendation Systems: Designed and deployed a shipment recommendation model using Scikit-learn to match user preferences with historical attributes, achieving an average response latency of 600ms. • Developed robust ETL pipelines in Python for clustering geospatial data into map layers, optimizing data storage by 98% and improving the customer experience in the search system.	
	Blockchain Engineer - Technical Consultant	01/2026 - 12/2026
	Bogotá, Cundinamarca	
	• Architecture design & proof of concept: Led the R&D of a proof of concept evaluating the implementation of distributed ledger technology (DLT) within a traditional banking ecosystem, focusing on secure ledger trust lists and real-world asset tokenization. • Technical & Strategic Documentation: Authored feasibility reports and presented technical findings to bank stakeholders, ensuring strict compliance with financial sector confidentiality agreements (NDA) while defining the infrastructure requirements needed to implement and scale the project.	
	Uninorte - Assistant Professor	07/2019 - 07/2020
	Barranquilla, Atlántico	
	• Instructed classes for digital signal processing using Matlab and Python focused on UI design using the Qt framework with an average score of 4.7/5 in the professor's review. • Laboratory assistant for the course of digital electronics design with DC-AC prototypes using the Arduino UNO board with an average score of 4.3/5 in the professor's review.	
	Psychology PhD Thesis Study - Data Analyst	06/2019 - 08/2019
	Barranquilla, Atlántico	
	• Designed and Analysis of experiments for an instrument that measures the falsehood or credibility of stories for children between 8 and 11 years old. • Data Analysis & NLP: Identified linguistic descriptors for the study group by applying Natural Language Processing (NLP) techniques using NLTK and Pandas, and extracted patterns based on neuropsychological knowledge using Machine Learning classification algorithms with Scikit-learn.	
	MadnessLab - Backend Developer	06/2019 - 12/2019
	Barranquilla, Atlántico	
	• Machine Learning applied to QoS: Generated sensitive analytical reports for user and operator metrics, improving the quality of service by 20% through NLP and Machine Learning techniques implemented in Python and Keras/PyTorch. ELT pipelines for processing call audio files recorded for credit institutions or financial companies with support for over 10 FFmpeg codecs.	

EXPERIENCE
ACADEMIC

Quatum Computing Framework - Python Developer Remote, Mexico	08/2021 - 10/2021
• Tutoring and subsidiary in Python and Numpy for the thesis modelling of a system/ environment in the framework of deeper entropy rise (SEA-QT) using random matrixes.	
Autonomous Surface Vessel - Backend Developer Barranquilla, Atlántico	01/2018 - 12/2018
• Developed a prototype related to Autonomous Surface Boats using a GPS receiver for measuring the position of the boat, a compass and other sensors and actuators for navigation using Python and Raspberry Pi platform. • Search algorithm using spherical geometry for the navigation with an error of 2 meters, leading to an acceptable error due to low-cost hardware limitations.	
AISMAP - Frontend Developer Barranquilla, Atlántico	01/2018 - 06/2018
• Build the UI for a mobile application on the Automatic Identification System (AIS) used by ships and vessel traffic services. • Accomplished time latency for the prototype close to 5s for transmitting data in real-time. • The AIVDM / AIVDO communication protocol was used to collect and integrate information from publicly available sources and vessels using the AIS protocol.	
Trucking Tracking - Full-Stack Developer Barranquilla, Atlántico	01/2018 - 06/2018
• Designed the UI for the trucks in transit with historical and real-time locations with an average response of 200ms using Javascript and HTML5 and CSS3. • Deployed the web application using Amazon Elastic Cloud Computing service EC2 for the server and the Amazon Relational Database Service for storage.	

PUBLICATIONS

International Journal on Communications Antenna and Propagation (IRECAP)	06/2019 - 12/2019
Obstacles, Speed and Spreading Factor: Insights in LoRa Mobile Performance.	

COURSES &
CERTIFICATES

Astronomer Certification DAG Authoring for Apache Airflow 3 • DAG and DAG authoring • ETL • Apache Airflow.	10/2025
Astronomer Certification for Apache Airflow 3 Fundamentals • Data pipelines • Orchestation and scheduling • Apache Airflow and Dags	10/2025
Introduction to Kubernetes • Microservices • Minikube • Volume Management • ConfigMaps • Container Orchestation • Secrets • Ingress	02/2025
Google Project Management Professional Certificate • Agile project management • Budgeting and procurement • Risk management • Effective Communication • Project scoping and planning • Stakeholder management • Work management tools • Process improvement	01/2025
API Academy Certification Program • API Designer. • API Security Architect. • API Product Manager.	08/2020 - 05/2021
Convolutional Networks for Visual Recognition • CS231n.	06/2018 - 08/2018
International Test of English Proficiency ITEP • CEFR Level B2.	04/2018 - 06/2018

EDUCATION

MSC on Electronics Engineering Universidad del Norte Identification of expressive descriptors for style extraction in music analysis using linear and non linear models.	06/2019 - 04/2023
BSC on Electronics Engineering Universidad del Norte System for the measurement of musical similarity, using expressive markers considering the acoustic intensity and temporal metrics.	06/2014 - 01/2019